JavaScript String Methods (Part 3)

This section covers more useful string methods:

- replace()
- replaceAll()
- split()
- repeat()
- charAt()
- indexOf()
- lastIndexOf()

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1. replace()

Definition

The `replace()` method replaces the **first occurrence** of a specified value in a string.

It returns a **new string** without changing the original string.

Syntax

```
string.replace(searchValue, newValue)
```

Example

```
let text = "I love Java. Java is easy.";

let result = text.replace("Java", "JavaScript");

console.log(result);
```

OUTPUT

```
I love JavaScript. Java is easy.
```

Only the **first** `"Java"` is replaced.

Original String

```
console.log(text);
```

Output

```
I love Java. Java is easy.
```

Real Project Example

Replace spaces with underscores.

```
let fileName = "My Resume.pdf";

console.log(fileName.replace(" ", "_"));
```

Output

```
My_Resume.pdf
```

---

2. replaceAll()

Definition

The `replaceAll()` method replaces **all occurrences** of a value.

Syntax

```
string.replaceAll(searchValue, newValue)
```

Example

```
let text = "Java Java Java";

console.log(text.replaceAll("Java", "JS"));
```

OUTPUT

```
JS JS JS
```

Real Project Example

```
let message = "Hello User. Hello Admin.";

console.log(message.replaceAll("Hello", "Welcome"));
```

Output

```
Welcome User. Welcome Admin.
```

Difference Between replace() and replaceAll()

replace()	replaceAll()
Replaces first match only	Replaces every match
More common	Useful for multiple replacements

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3. split()

Definition

The `split()` method converts a string into an array.

Syntax

```
string.split(separator)
```

Example

```
let text = "HTML CSS JavaScript";

console.log(text.split(" "));
```

OUTPUT

```
["HTML","CSS","JavaScript"]
```

Split by Comma

```
let fruits = "Apple,Mango,Orange";

console.log(fruits.split(","));
```

Output

```
["Apple","Mango","Orange"]
```

Split Every Character

```
let word = "Code";

console.log(word.split(""));
```

Output

```
["C","o","d","e"]
```

Real Project Example

Tags

```
let skills = "HTML,CSS,JavaScript,React";

let skillArray = skills.split(",");

console.log(skillArray);
```

---

4. repeat()

Definition

The `repeat()` method repeats a string a specified number of times.

Syntax

```
string.repeat(count)
```

Example

```
let star = "*";

console.log(star.repeat(10));
```

OUTPUT

```
**********
```

Example

```
let text = "Hi ";

console.log(text.repeat(3));
```

Output

```
Hi Hi Hi
```

Real Project Example

```
console.log("-".repeat(40));
```

Output

```
----------------------------------------
```

Useful for console formatting.

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5. charAt()

Definition

The `charAt()` method returns the character at a specified index.

Syntax

```
string.charAt(index)
```

Example

```
let language = "JavaScript";

console.log(language.charAt(0));
```

OUTPUT

```
J
```

Example

```
console.log(language.charAt(4));
```

Output

```
s
```

Another Way

```
console.log(language[4]);
```

Output

```
s
```

Modern JavaScript usually prefers bracket notation.

---

6. indexOf()

Definition

The `indexOf()` method returns the **first index** of a specified value.

If the value is not found, it returns **-1**.

Syntax

```
string.indexOf(searchValue)
```

Example

```
let language = "JavaScript";

console.log(language.indexOf("Script"));
```

OUTPUT

```
4
```

Example

```
console.log(language.indexOf("Python"));
```

Output

```
-1
```

Real Project Example

```
let email = "admin@gmail.com";

if(email.indexOf("@") !== -1){
    console.log("Valid Email");
}
```

Output

```
Valid Email
```

7. lastIndexOf()

Definition

The `lastIndexOf()` method returns the **last occurrence** of a value.

Syntax

```
string.lastIndexOf(searchValue)
```

Example

```
let text = "Java Java Java";

console.log(text.lastIndexOf("Java"));
```

OUTPUT

```
10
```

Example

```
let sentence = "one two one";

console.log(sentence.lastIndexOf("one"));
```

Output

```
8
```

Real Project Example

```
let file = "photo.backup.jpg";

console.log(file.lastIndexOf("."));
```

Output

```
12
```

Useful for finding file extensions.

Summary Table

Method	Purpose	Returns
replace()	Replace first match	String
replaceAll()	Replace all matches	String
split()	Convert string to array	Array
repeat()	Repeat string	String
charAt()	Character at index	String
indexOf()	First occurrence	Number
lastIndexOf()	Last occurrence	Number

Most Frequently Used

⭐⭐⭐⭐⭐

- split()
- replace()
- indexOf()

⭐⭐⭐⭐

- replaceAll()
- charAt()

⭐⭐⭐

- repeat()
- lastIndexOf()

Common Mistakes

REPLACE()

❌ Expecting all matches to be replaced.

```
"Java Java".replace("Java","JS")
```

Output

```
JS Java
```

SPLIT()

Using the wrong separator.

```
"HTML CSS".split(",")
```

Output

```
["HTML CSS"]
```

INDEXOF()

Always check for `-1`.

```
if(text.indexOf("Java") !== -1){
    console.log("Found");
}
```

Interview Questions

Q1. What is the difference between replace() and replaceAll()?

replace()	replaceAll()
Replaces first occurrence	Replaces all occurrences

Q2. Which method converts a string into an array?

```
split()
```

Q3. Which method repeats a string?

```
repeat()
```

Q4. Which method returns the first occurrence?

```
indexOf()
```

Q5. Which method returns the last occurrence?

```
lastIndexOf()
```

Q6. Which method returns a character at a specific position?

```
charAt()
```

JavaScript Strings (Part 4)

This section covers:

- Reverse a String (3 Methods)
- Real Project Examples
- Common Mistakes
- Interview Questions
- Summary Table

Reverse a String

What is String Reversal?

Reversing a string means changing the order of its characters from the last character to the first.

Example

```
JavaScript

↓

tpircSavaJ
```

Method 1 (Most Popular)

split() + reverse() + join()

EXPLANATION

Step 1

Convert the string into an array.

```
"Java"

↓

["J","a","v","a"]
```

Step 2

Reverse the array.

```
["a","v","a","J"]
```

Step 3

Join the array back into a string.

```
"avaJ"
```

Code

```
let text = "JavaScript";

let reverseText = text.split("").reverse().join("");

console.log(reverseText);
```

OUTPUT

```
tpircSavaJ
```

Step by Step

```
let text = "Code";

console.log(text.split(""));
```

Output

```
["C","o","d","e"]
```

```
console.log(text.split("").reverse());
```

Output

```
["e","d","o","C"]
```

```
console.log(text.split("").reverse().join(""));
```

Output

```
edoC
```

Advantages

✅ Easy

✅ Short

✅ Most commonly used

Method 2 (Using for Loop)

Explanation

Start from the last character.

Move toward the first character.

Store every character inside a new string.

Code

```
let text = "JavaScript";

let reverse = "";

for(let i = text.length - 1; i >= 0; i--){

    reverse += text[i];

}

console.log(reverse);
```

OUTPUT

```
tpircSavaJ
```

Visualization

```
Java

Index

0 1 2 3

J a v a

↓

Start

3

↓

a

↓

2

↓

v

↓

1

↓

a

↓

0
```

```
↓

J

↓

avaJ
```

Advantages

Good for learning loops.

Frequently asked in interviews.

Method 3 (Using for...of Loop)

Code

```
let text = "JavaScript";

let reverse = "";

for(const letter of text){

    reverse = letter + reverse;

}

console.log(reverse);
```

OUTPUT

```
tpircSavaJ
```

Explanation

Iteration 1

```
J
```

Reverse

```
J
```

Iteration 2

```
a + J

↓

aJ
```

Iteration 3

```
v + aJ

↓

vaJ
```

Continue...

Final

```
tpircSavaJ
```

Which Method is Best?

Method	Easy	Interview	Real Project
split().reverse().join()	⭐⭐⭐⭐⭐	⭐⭐⭐	⭐⭐⭐⭐⭐
for Loop	⭐⭐⭐⭐	⭐⭐⭐⭐⭐	⭐⭐⭐⭐
for...of	⭐⭐⭐⭐	⭐⭐⭐⭐	⭐⭐⭐

Real Project Examples

Username Validation

```
let username = "Rokon";

if(username.length >= 5){

    console.log("Valid Username");

}else{

    console.log("Too Short");

}
```

Output

```
Valid Username
```

Email Validation

```
let email = "admin@gmail.com";

if(email.includes("@")){

    console.log("Valid Email");

}
```

Output

```
Valid Email
```

Password Validation

```
let password = "abc12345";

if(password.length >=8){

    console.log("Strong Password");

}
```

Output

```
Strong Password
```

Search Feature

```
let product = "JavaScript Course";

if(product.toLowerCase().includes("javascript")){

    console.log("Found");

}
```

Output

```
Found
```

Remove Extra Spaces

```
let name = "     Rokon     ";

console.log(name.trim());
```

Output

```
Rokon
```

File Extension Check

```
let file = "resume.pdf";

console.log(file.endsWith(".pdf"));
```

Output

```
true
```

Website URL Check

```
let url = "https://google.com";

console.log(url.startsWith("https"));
```

Output

```
true
```

Display User Name

```
let firstName = "MD";
let lastName = "Rokon";

console.log(`${firstName} ${lastName}`);
```

Output

```
MD Rokon
```

Common Mistakes

Forgetting Quotes

❌

```
let text = JavaScript;
```

Error

Wrong Index

```
let text = "Java";

console.log(text[10]);
```

Output

```
undefined
```

Expecting replace() to Replace Everything

❌

```
let text = "Java Java";

console.log(text.replace("Java","JS"));
```

Output

```
JS Java
```

Use

```
replaceAll()
```

Forgetting trim()

```
let username = "   admin   ";

console.log(username === "admin");
```

Output

```
false
```

Correct

```
username.trim()
```

Using == Instead of ===

❌

```
if(name == "Rokon")
```

Prefer

```
if(name === "Rokon")
```

Interview Questions

What is a String?

A string is a sequence of characters used to represent text.

Are Strings Mutable?

No.

Strings are immutable.

Which method converts a string into an array?

```
split()
```

Which method removes spaces?

```
trim()
```

Which method converts to lowercase?

```
toLowerCase()
```

Which method checks whether text exists?

```
includes()
```

Which method extracts part of a string?

```
slice()
```

Which method joins strings?

```
concat()
```

How do you reverse a string?

Most common solution

```
text.split("").reverse().join("")
```

Which method returns the total number of characters?

```
length
```

Difference Between slice() and substring()

slice()	substring()
Supports negative index	Doesn't support negative index

Summary Table

Topic	Method	Most Used
Convert Lowercase	toLowerCase()	⭐⭐⭐⭐⭐
Convert Uppercase	toUpperCase()	⭐⭐⭐⭐
Remove Spaces	trim()	⭐⭐⭐⭐⭐
Extract String	slice()	⭐⭐⭐⭐⭐
Extract String	substring()	⭐⭐⭐
Join Strings	concat()	⭐⭐⭐
Search Text	includes()	⭐⭐⭐⭐⭐
Check Start	startsWith()	⭐⭐⭐⭐
Check End	endsWith()	⭐⭐⭐⭐
Replace Text	replace()	⭐⭐⭐⭐⭐
Replace All	replaceAll()	⭐⭐⭐⭐
Convert to Array	split()	⭐⭐⭐⭐⭐
Repeat Text	repeat()	⭐⭐
Character Access	charAt()	⭐⭐⭐
First Position	indexOf()	⭐⭐⭐⭐
Last Position	lastIndexOf()	⭐⭐⭐
Reverse String	split().reverse().join()	⭐⭐⭐⭐⭐

Final Notes

For JavaScript, React, and Node.js development, focus on mastering these methods:

- length
- toLowerCase()

- toUpperCase()
- trim()
- slice()
- includes()
- replace()
- split()
- startsWith()
- endsWith()
- indexOf()

These methods are used almost every day in real-world projects and are commonly asked in JavaScript interviews.